Authentication vs. Authorization

Here's a comparison of authentication and authorization:

Aspect	Authentication	Authorization
Definition	The process of verifying who a user is.	The process of verifying what access a user has.
Focus	Identity verification.	Access rights and privileges.
Example	Entering a username and password.	Checking if a user can access a specific resource, like a file or database.
How It Works	Typically involves passwords, biometrics, OTPs, etc.	Involves settings, roles, and permissions that define what resources a user can access.
Tools/Methods	Login forms, OTPs, biometric scanners.	Access control lists, role-based access control.
Order in Process	Comes first in the security process.	Follows after authentication is successful.
Key Concern	Verifying user identity is genuine.	Managing user permissions and access levels.
Frequency	Typically occurs once at the beginning of a session.	Can occur multiple times, whenever a user requests access to a resource.
Dependence	Independent process, can exist without authorization in some systems.	Requires authentication as a prerequisite.